Milind Saxena

Data Analyst · SQL · Python · Power BI · Databricks · Excel

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SUMMARY

Detail-oriented data analyst with an Honours B.A. in Computer Science & Economics (McMaster University, Dean's Honour List). I turn large, messy datasets into decisions: SQL, Python and Spark for data cleaning, ETL and statistical analysis; A/B testing, forecasting and machine learning; Power BI dashboards and KPI reporting; and clear write-ups for technical and non-technical stakeholders. Every analysis I ship includes automated data-quality checks.

EDUCATION & CERTIFICATIONS

Honours B.A., Computer Science & Economics, McMaster University Hamilton, ON · Nov 2023
Dean's Honour List (2019)
Certifications: Databricks Certified Data Engineer Associate (Nov 2026) · Microsoft Certified: Power BI Data Analyst Associate, PL-300 (Oct 2026)

TECHNICAL SKILLS

Core: SQL (joins, CTEs, window functions, aggregations) · Python (pandas, scikit-learn) · Power BI (DAX, data modelling, dashboards) · Databricks (Spark SQL, PySpark, Delta Lake) · Excel · statistics (A/B testing, regression, hypothesis testing, forecasting)
Also used: statsmodels, scipy, DuckDB, Tableau, matplotlib, Jupyter, REST APIs, Git & GitHub, unit testing
Practices: data cleaning, ETL / data pipelines, data validation and quality checks, KPI reporting, experiment design, trend analysis, data visualization, data storytelling, documentation

DATA ANALYTICS PROJECTS

Toronto Subway Delays: why delays nearly doubled since 2014 SQL · Python · Statistics

- Built a Python and SQL (DuckDB) pipeline that cleaned 267,571 hand-typed TTC incident records (111 line spellings to 5, 2,221 locations to 75 stations via alias and fuzzy matching), with 18 automated data-quality checks.
- Found delay minutes rose 85% and 76% of the increase came from passenger incidents, not equipment; a difference-in-differences test showed Line 1's new signal system halved its signal delays.

Mobile Game A/B Test: catching a broken experiment Python · SQL · A/B Testing

- Analysed a 90,189-player A/B test: a sample ratio mismatch check and an A/A segment found 823 extra low-activity users in one group ($p = 0.0005$), which more than halved the true retention effect once corrected.
- Built a reusable statistics toolkit (z-tests, Bayesian comparison, power and sample-size analysis) with 13 unit tests, and wrote the decision memo: keep the current design and re-run with 2.4× the sample.

Customer Churn: who is worth saving? Python · scikit-learn · SQL

- Built and calibrated churn models (logistic regression, random forest, gradient boosting; AUC 0.84) on 7,043 customers with 5-fold cross-validation and permutation importance.
- Replaced the default 50% threshold with expected-profit targeting (churn risk × customer value), raising simulated campaign profit from $85k to $92k and winning in all 30 cost scenarios; delivered a ranked target list with reason codes.

UPI vs. Cash: why India still holds record cash SQL · Python · Forecasting

- Built an ETL pipeline and 7 SQL data models over RBI and World Bank data with 15 automated data-quality checks; showed cash held near 12% of GDP while UPI grew from 37% to 91% of GDP.
- Forecast monthly cash demand with Holt-Winters (1.1% average error in a 25-period rolling backtest): ₹48.1 lakh crore by May 2027, so banks should plan cash logistics for growth, not decline.

Additional projects github.com/BoredMongoose

- **Express Entry draws** (SQL, Python, forecasting): modelled all 448 IRCC draws with 11 data-quality checks (one caught an error in IRCC's own table); a backtested model predicts the next cutoff with a 2.4-point average error.
- **Delhi air pollution** (Python, statistics, APIs): 600,000+ satellite fire records and a weather-normalization model attributing ~60% of early-November PM2.5 to crop-fire smoke.
- **Moneyball IPL** (SQL, pandas): a "runs added" KPI from 290,000+ ball-by-ball records with window functions; fuzzy-matched auction names; price explained only ~10% of player impact.

EXPERIENCE

Customer Service Representative, Domino's Pizza Hamilton, ON · Apr 2023 – Aug 2023

- Handled high-volume orders and customer inquiries accurately against tight deadlines in a fast-paced team.
- Trained 7 new team members on store operations, policies and service standards.